Literature-Implied One-Dimensional Subcase for arXiv:1501.07036

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2026-06-19

Status

Status: literature-implied answer, partial subcase.

The source paper is E. Pernecka and R. J. Smith, *The metric approximation property and Lipschitz-free spaces over subsets of* $\mathbb{R}^N$, arXiv:1501.07036. After Corollary 1.2, the authors ask whether $\mathcal{F}(M)$ has the metric approximation property for all subsets $M \subset \mathbb{R}^N$.

The supporting paper is A. Godard, *Tree metrics and their Lipschitz-free spaces*, arXiv:0904.3178.

Recorded Subcase

For every subset $M \subset \mathbb{R}$, equipped with the inherited Euclidean metric, the Lipschitz-free space $\mathcal{F}(M)$ has the metric approximation property.

More generally, if A is a subset of an $\mathbb{R}$-tree T such that $\mathrm{Br}(T) \subset \overline{A}$, then $\mathcal{F}(A)$ has the metric approximation property.

Identification

Godard's Theorem 3.2 states that, for an $\mathbb{R}$-tree T and a subset $A \subset T$ with $\mathrm{Br}(T) \subset \overline{A}$, the free space $\mathcal{F}(A)$ is isometric to $L_1(\mu_{\overline{A}})$, where $\mu_{\overline{A}}$ is the explicit length-plus-gap measure associated with $\overline{A}$. In the special case $T = \mathbb{R}$, the tree has no branching points, so the hypothesis is automatic for every subset $M \subset \mathbb{R}$. Therefore $\mathcal{F}(M)$ is isometric to an L_1 space.

Every $L_1(\mu)$ space has MAP: finite measurable partitions yield norm-one conditional expectations with finite-dimensional ranges, and these expectations converge strongly to the identity along the directed set of finite subalgebras. Hence $\mathcal{F}(M)$ has MAP for all $M \subset \mathbb{R}$.

Scope

This is not a full solution of the Pernecka–Smith question. It covers the complete one-dimensional case $N = 1$, and the same reasoning covers the tree-metric subcase above. The arbitrary subset problem in $\mathbb{R}^N$ for $N \geq 2$ remains open in this packet.

References

- E. Pernecka and R. J. Smith, *The metric approximation property and Lipschitz-free spaces over subsets of* $\mathbb{R}^N$, arXiv:1501.07036.

- A. Godard, *Tree metrics and their Lipschitz-free spaces*, arXiv:0904.3178.